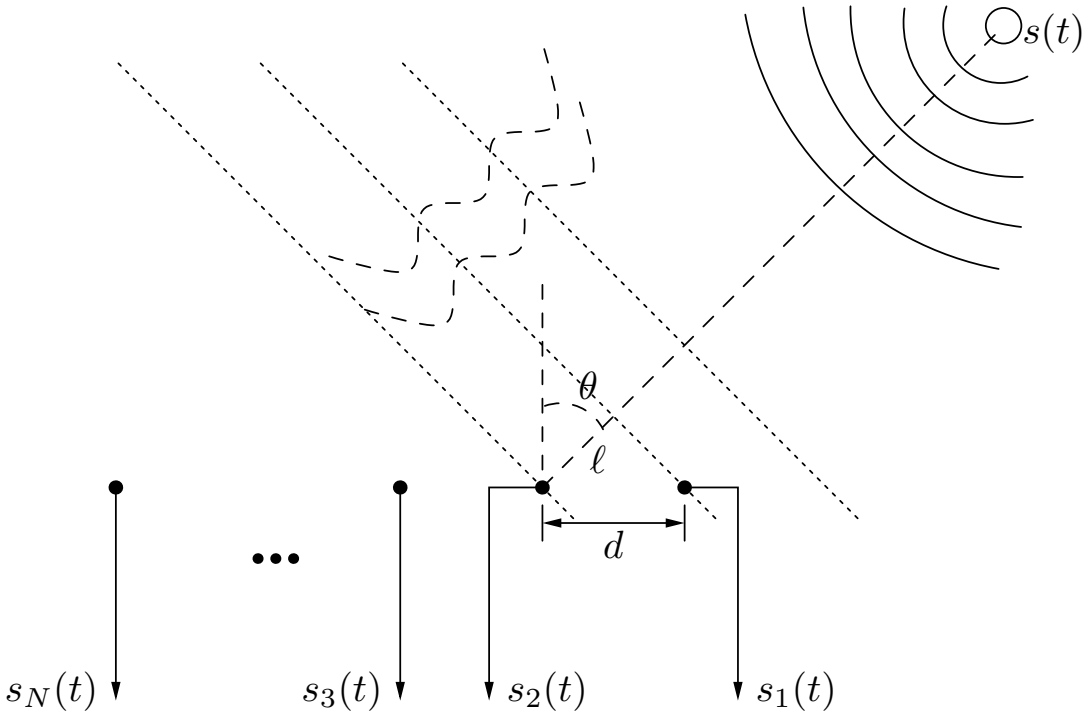
Source Localization Using a Single Array of Sensors in Nearfield Configuration, and Application to Active Noise Cancellation

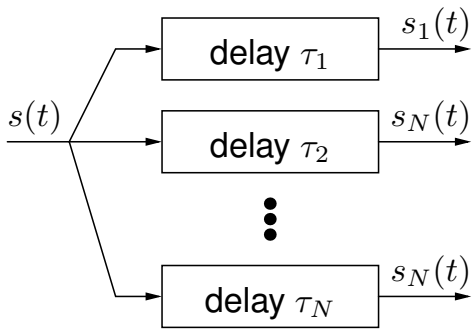
Todd K. Moon, Mohammad Shekaramiz, and Jacob H. Gunther

Information Dynamics Laboratory, ECE Dept., Utah State University

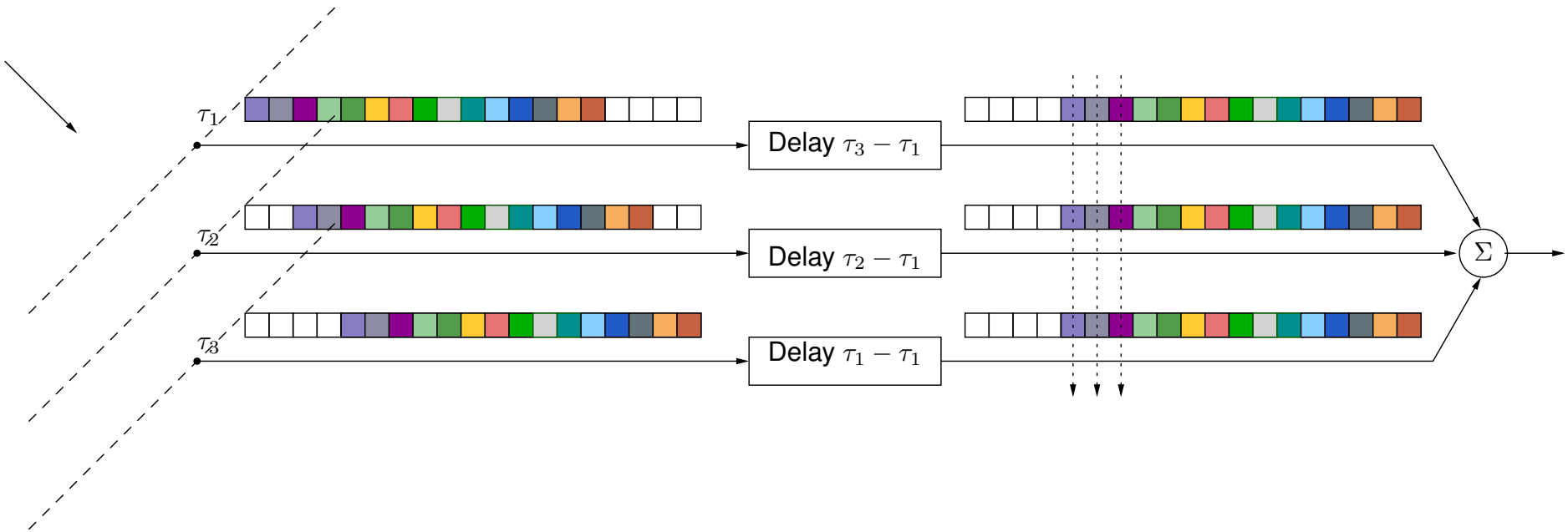
May 2024

Linear Array, far field





Basic Processing Concept: Delay and Add



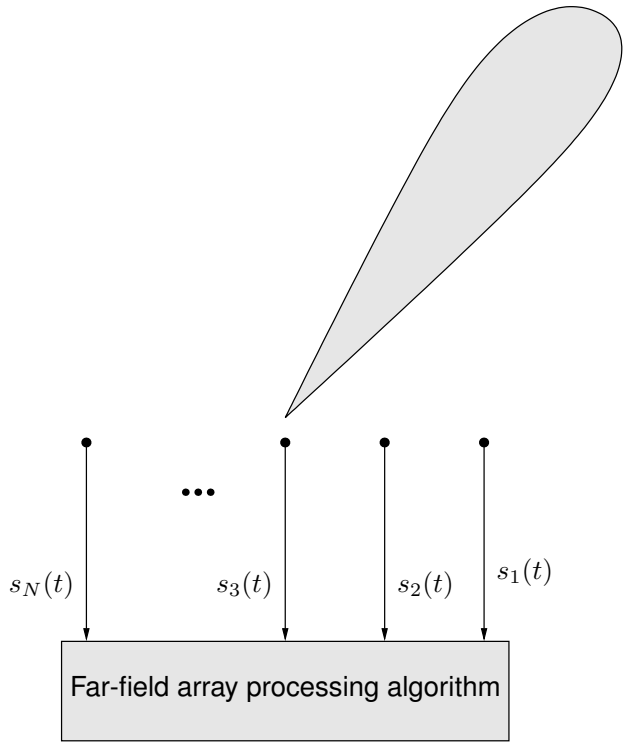
Simplifications for narrowband, farfield, linear arrays

Narrowband signal:

$s(t) = m(t)e^{j2\pi f_c t}$ $m(t)$ slowly changing

$$\begin{aligned} s(t-\tau) &= m(t-\tau)e^{j2\pi f_c(t-\tau)} \\ &\approx m(t)e^{j2\pi f_c t}e^{-j2\pi f_c \tau} \\ &= s(t)e^{-j2\pi f_c \tau} \end{aligned}$$

Delays are represented by phase.



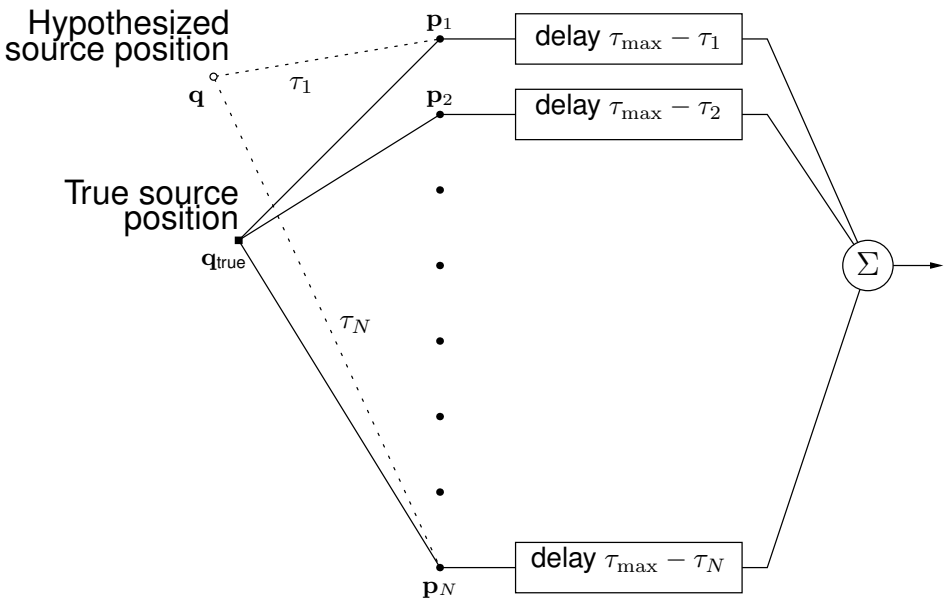
Linear, farfield array:

Delay difference: $\tau_2 - \tau_1 = \frac{d\sin\theta}{c} \triangleq \Delta\tau$,
Constant for all adjacent sensors.
Farfield array processing algorithms determine the **direction** to the source.

Removing the simplifying assumptions

- Not necessarily narrowband. Implement the delays explicitly
- Not farfield: Wavefronts are assumed to be spherical.
- Not linear: Can apply to arbitrarily positioned sensors.

Delay-sum combining



- There has been substantial work on the near-field problem, but few visual examples of performance.
- Increasing processing speeds make simulating delays more feasible, avoiding analytical approximations examined in the literature.

Creating the signal delays from sampled signals

$$s(t-\tau) = \sum_{\ell=-\infty}^{\infty} s(\ell T)\,\mathrm{sinc}(2B((t-\tau)-\ell T))$$

Sampled delayed signal, truncated

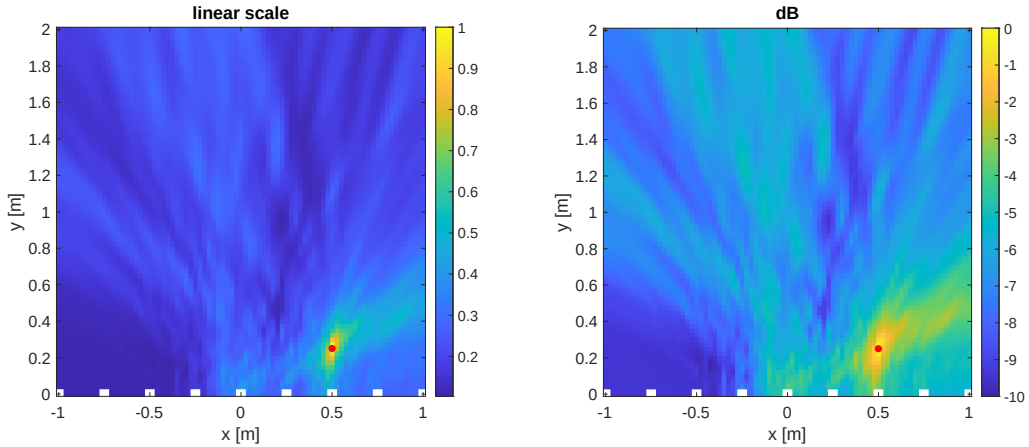
$$s(nT-\tau) = \sum_{\ell=-P}^{P} s(\ell T)\,\mathrm{sinc}(2B((nT-\tau)-\ell T))$$

Impulse response:

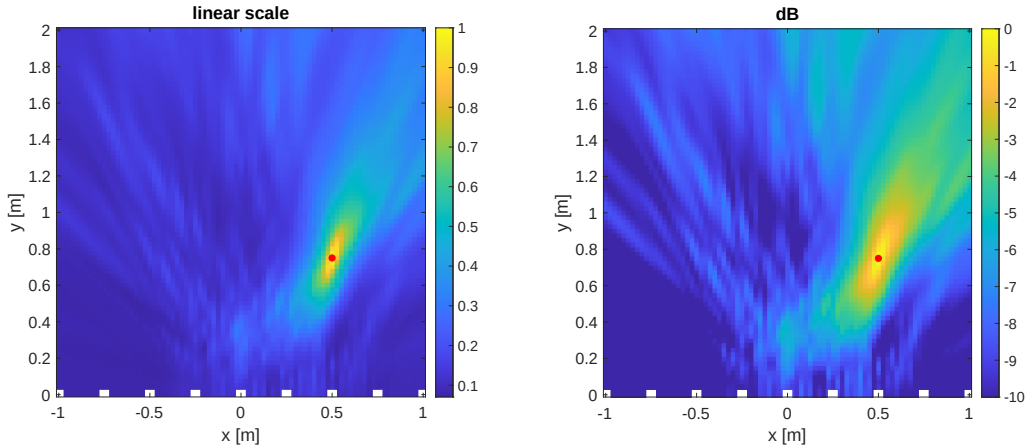
$$h_\tau(n) = \mathrm{sinc}(2B(nT-\tau)), \quad n = -P, \ldots, P$$

(Delay to make it causal.)

Some Examples — Linear Array



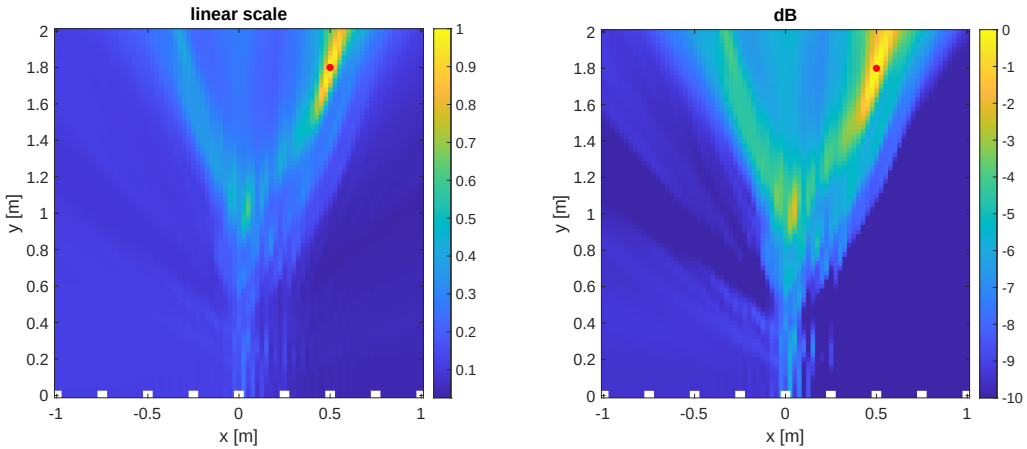
(a) True source position = (0.5, 0.25)



(b) True source position = (0.5, 0.75)

The explicit delay-sum combining provides good spatial localization.

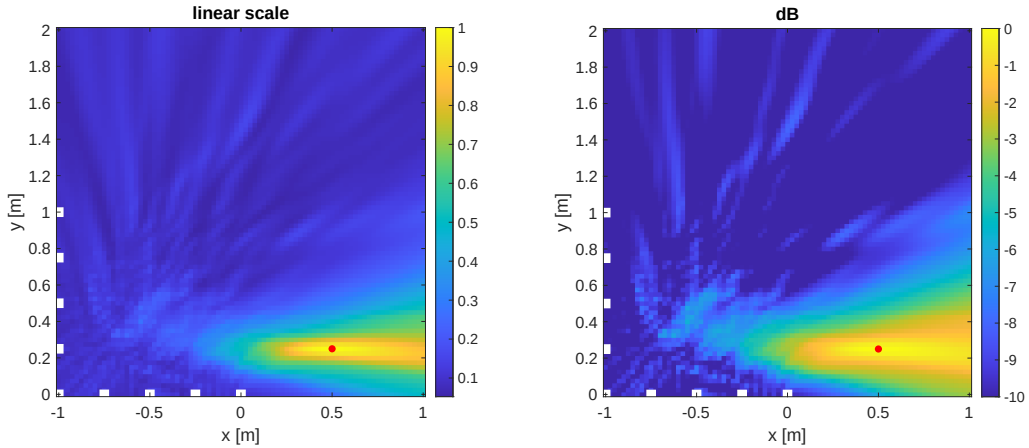
Some Examples — Linear Array



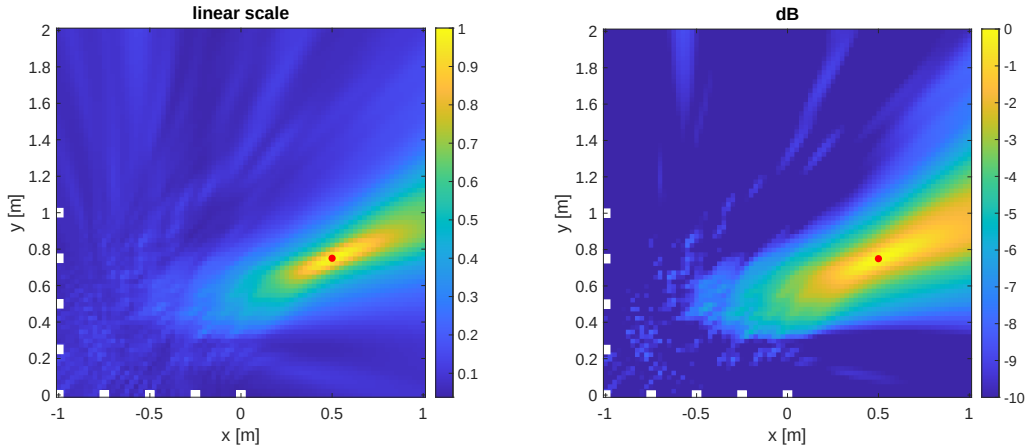
(c) True source position = (0.5, 1.8)

As the source moves away from the sensor array, the region becomes more beam-like.

Some Examples — L-shaped array

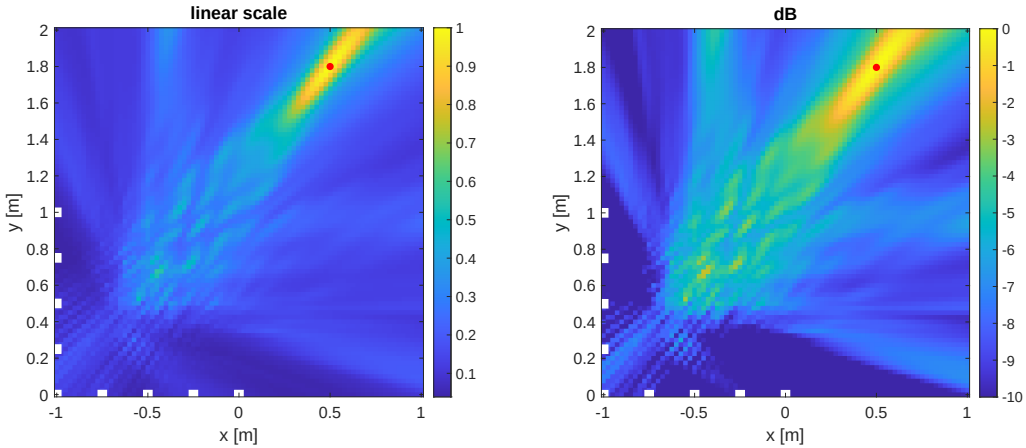


(d) True source position = (0.5, 0.25)



(e) True position = (0.5, 0.75)

Some Examples — L-shaped array



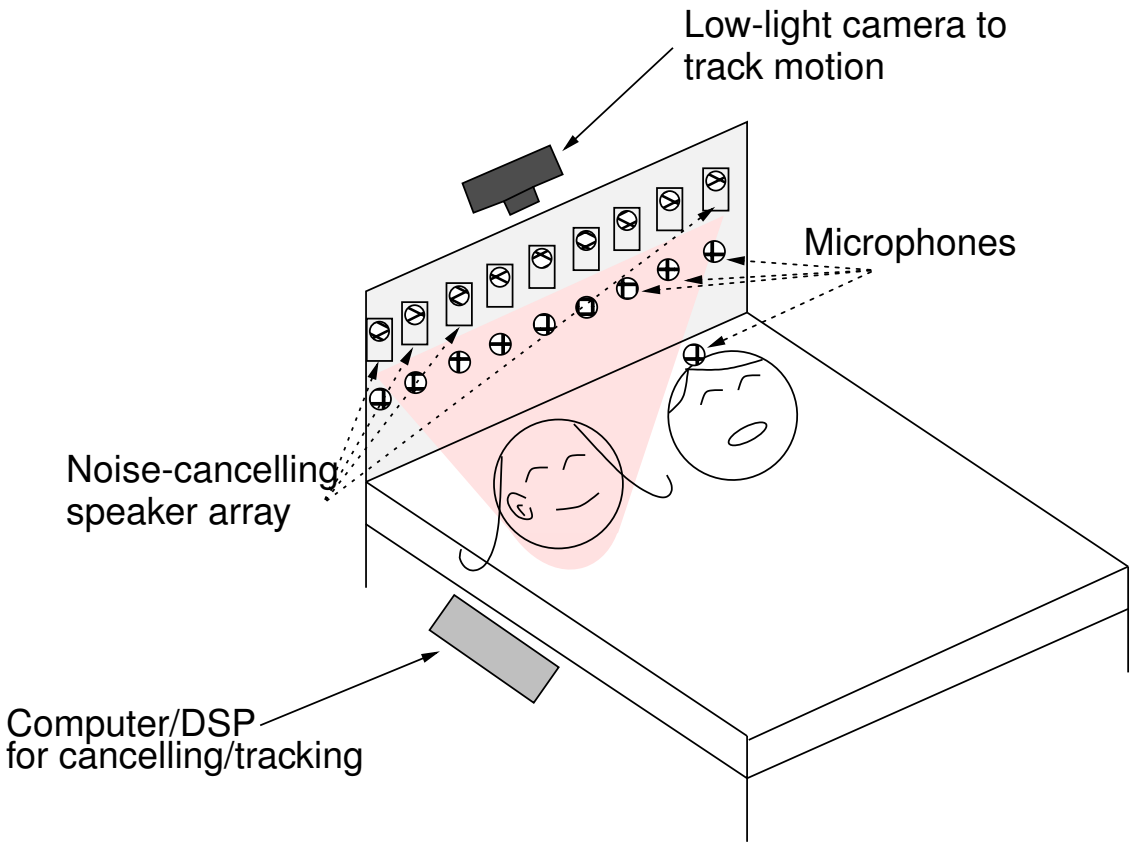
(f) True source position = (0.5, 1.8)

Test setup

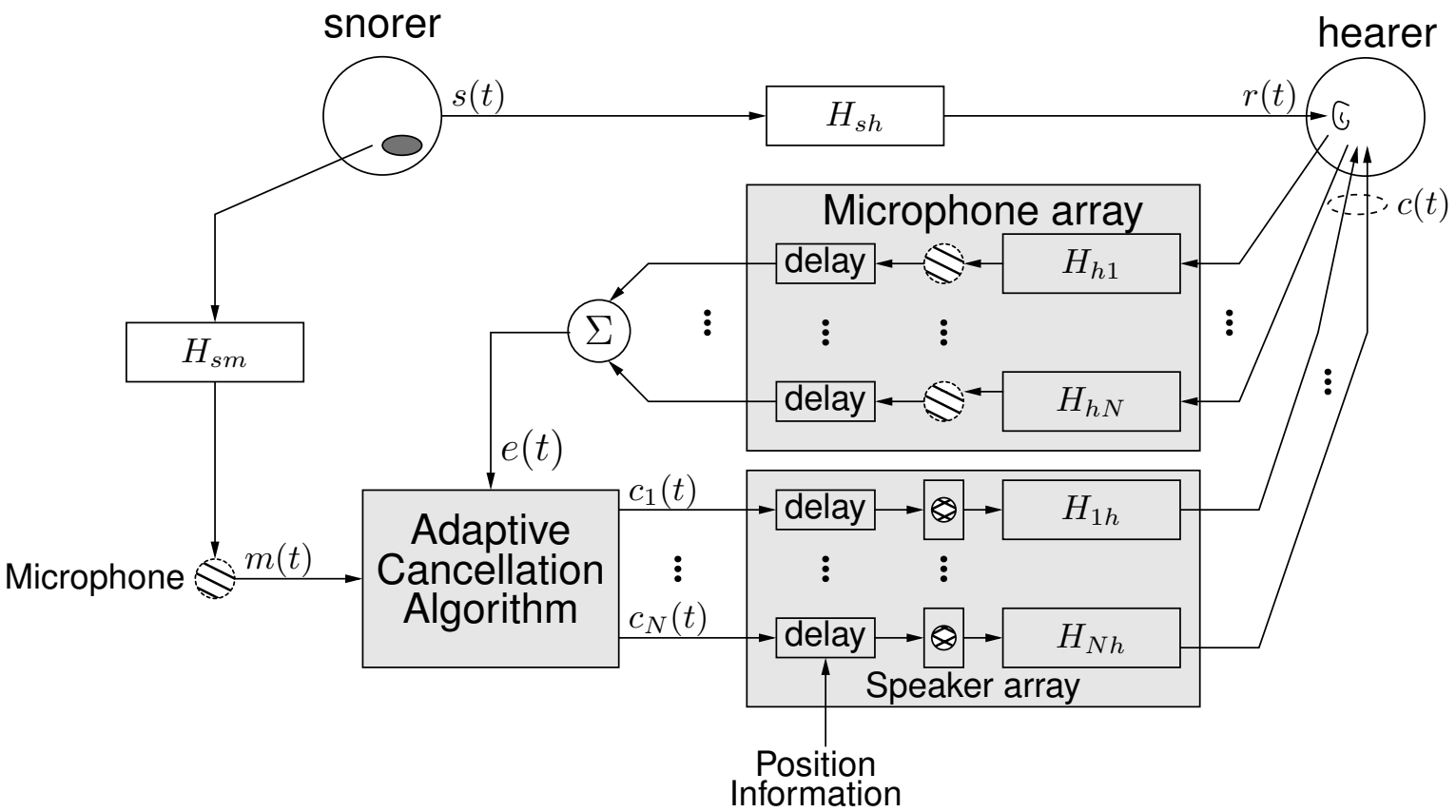


Why might we care about near-field beamforming on broadband signals?

- Up to a quarter of the population is affected by snoring.
- Impacts: strong statistical impact on marital state and quality of life.
- Elimination of snoring was associated with an improvement in the quality of partners' sleep.



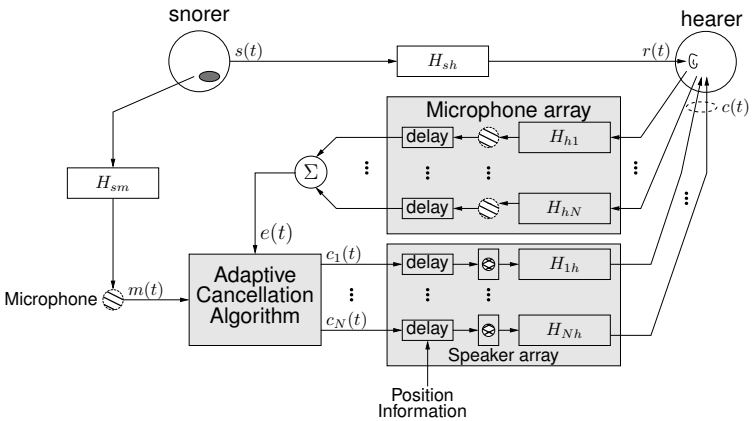
Block diagram



Adaptive Cancellation

- Cancellation signal $c[n]$. (Steered by array).
- Canceller: $G(z)$.
- Canceller Error

$$
\begin{aligned}
e[n] &= c[n] - r[n] \\
&= \sum_{i=0}^{M-1} c[i]g[n-i] - r[n] \\
&= \mathbf{c}[n]^T \mathbf{g}[n] - r[n]
\end{aligned}
$$



- Minimize $J(\mathbf{g}) = E[e[n]^2]$.
- LMS-type update:

$$\mathbf{g}^{[n+1]} = \mathbf{g}^{[n]} - \mu \mathbf{c}[n]e[n]$$

Summary

- Near-field arrays provide spatial localization, not merely directional localization.
- The processing is enabled by increases in signal processing capability.
- This opens the possibility of local noise cancellation applications.
- Snoring cancelling is proposed as one such possibility.